

Experiment 1

Preparation of analytical reagents and calibration standards

Purpose

1. to prepare analytical and calibration standards
2. to learn how to determine the concentration of the diluted solutions

Background

Analytical methods usually contain separate section for reagents and calibration standards. Analytical reagents are commonly used in classical methods such as barium chloride solution in gravimetric determination of sulfate and hydrochloric acid in determination of carbonates. Calibration standards are commonly used in instrumental methods since each instrument requires calibration curve before analysis. For example flame photometer is a spectroscopic method for sodium measurement by transferring it into a yellow radiation emits at 580 nm. This instrument requires calibration with a set of Na^+ aqueous solutions between zero and 100 ppm before being used.

Usually we buy analytical reagents as concentrated solutions; 37% (w/w) HCl, or solid salts. To prepare a dilute reagent, first we have to know how to calculate the molarity of the concentrated reagent and then how much we should take from it to prepare the dilute one. In the case of calibration standard, we have to know how much to take from the solid salt containing the tested element i.e Na in NaCl salt. The concentration of stock solution should be five times than the highest calibration standard. Calibration standards with lower concentration are prepared by subsequent (serial) dilution.

Calculation example:

1. How many mL's or grams of HCl (36.5 g/mol) solution (37% (W/W) with density of 1.19 g/mL are required to prepare 250 mL of 0.25 M HCl solution?

Solution

Assume that 100 g of HCl solution contains 37.0 g of HCl

$M = \text{number of moles of HCl} / \text{Volume solution (L)}$

$\text{Number of moles (n)} = \text{weight of HCl} / \text{formula weight HCl}$

$\text{Volume} = \text{mass} / \text{density}$

Using the dimensional analysis method:

$$0.25 \frac{\text{mol}}{\text{L}} * 0.25 \text{L} * 36.45 \frac{\text{g}}{\text{mol}} * \frac{100}{37} = 6.33 \text{g}$$

$$\text{volume} = \frac{\text{mass}}{\text{density}} = \frac{6.33}{1.19} = 5.3 \text{mL}$$

2. You have been asked to prepare 100 mL of 250 ppm Na^+ solution.

a) How many mgs of NaCl (58.5g/mol) you will take to prepare this solution?

b) How many mgs of Na_2SO_4 (142 g/ mol) you will take to prepare this solution ?

ppm= mg/L

Gravimetric factor (G.F.) = (F.wt sought/ F.wt source)*(mol ratio)

Solution for a) 1 mmol NaCl $\longrightarrow$ 1 mmol Na^+

$$250 \frac{\text{mg}}{\text{L}} * 0.1 \text{L} * \left(\frac{58.5}{23} * \frac{1}{1} \right) = 63 \text{mg NaCl}$$

Solution for b) 1 mmol Na_2SO_4 $\longrightarrow$ 2 mol Na^+

$$250 \frac{\text{mg}}{\text{L}} * 0.1 \text{L} * \left(\frac{142}{23} * \frac{1}{2} \right) = 77.17 \text{mg Na}_2\text{SO}_4$$

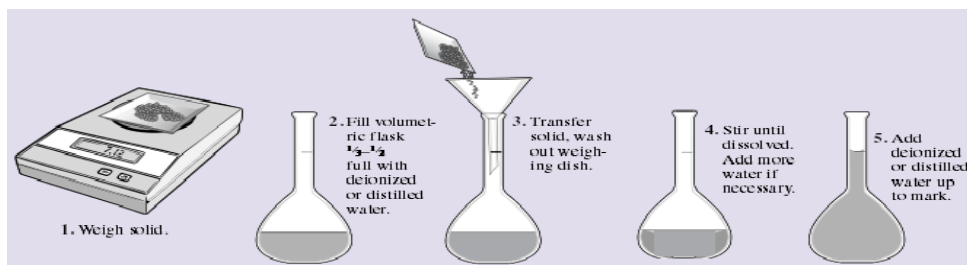
Procedure:

A. Preparation of 250 mL of 0.20 M solution of a strong acid:

1. Ask your instructor to assign to you an acid.
2. From the label on the bottle write the formula weight, density and % concentration into the notebook
3. Calculate the amount required
4. Weight the calculated amount ($\pm 0.01\text{g}$) in a weighing bottle.
5. Obtain a 250 mL volumetric flask, fill it to half with distilled water, add the acid carefully into the flask through a funnel, wash the weighing funnel and bottle several times with distilled water.
6. Complete the volume to the mark by distilled water.

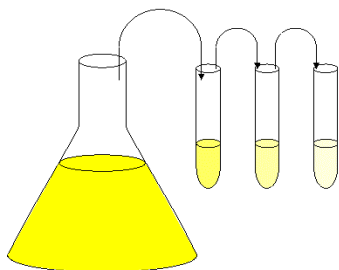
B. Preparation of 250 mL of 0.20 M solution of a strong base:

1. Ask your instructor to assign a base to you.
2. From the label on the bottle transfer the formula wt and % into your notebook.
3. Repeat steps 3 to 6 as in part A.



C. Preparation of 100 mL calibration solutions of 1.00, 5.00, 10.0, and 25.0 ppm of assigned element:

1. Ask your instructor to assign you a compound containing the assigned element.
2. From the label on the bottle transfer the formula .wt and % into your notebook
3. Calculate the amount required to prepare stock solution of 250 mL of 100 ppm solution.
4. Weight the calculated amount ($\pm 0.0001\text{g}$) in a weighing bottle.
5. Repeat steps 3 to 6 as in part A.
6. Prepare the four calibration standard from the stock solution by serial dilution.



Questions

- 1- How many mL's or grams of HCl (36.45g/mol) (37%) , density of 1.19g/mL , are required to prepare 250 mL of 0.25 M HCl solution?
- 2- How many grams of NaOH (40 g/mol) are required to prepare 300 mL of 0.079M?

Experiment (1) - Report sheet

A) Preparation of 250 mL of 0.20 M solution of a strong acid:

Name of acid: Molecular Formula:

Formula weight: Purity:

Calculations:

B) Preparation of 250 mL of 0.20 M solution of a strong base:

Name of base: Molecular Formula:

Formula weight: Purity:

Calculations:

C) Preparation of 100 mL calibration solutions of 1.00, 5.00, 10.0, and 25.0 ppm of element:

Assigned element:

Name of compound: Formula weight: Purity:

Calculations:

- The weight of **compound** required to prepare 250 mL of 100 ppm of element stock solution:

- The volume of 100 ppm element stock solution required to prepare each of the following:

250 mL of 1.00 ppm of element solution:

250 mL of 5.00 ppm of element solution:

250 mL of 10.0 ppm of element solution:

250 mL of 1.00 ppm of element solution:

Experiment 2

Statistical Evaluation of Analytical Results

Purpose

To learn some statistical calculation that could be useful in the evaluation of analytical results

Back ground

Analytical results usually reflect a situation from which a decision will be made such as in medicinal and environmental measurements. Thus these results will be subjected to certain rules before being accepted or rejected. Statistics give us tools to accept conclusion that have a high probability of being correct and to reject conclusion that do not. This experiment applies the basic statistical rules that are widely used in analytical chemistry. For an analytical method, usually we collect finite number of measurements X_i , where $i=1$ to n , where n is the number of replicates. From these measurements we calculate the average $\bar{x}$ (also called average) is the sum of measured values divided by n

$$\text{Average: } \bar{x} = \sum_i \frac{x_i}{n}$$

The standard deviation, s measures how closely the replicates are clustered about the average:

$$\text{Standard deviation: } S = \sqrt{\sum_i (x_i - \bar{x})^2 / n - 1}$$

Usually it's more meaningful to represent the percentage relative standard deviation known as the coefficient of variation (CV) rather than the absolute standard deviation:

$$\text{Coefficient of variation: } CV = \left(\frac{S}{\bar{x}} \right) * 100\%$$

From a limited number of measurements, we cannot find the true populations mean μ . The confidence interval is an expression stating that the true mean μ is likely to lie within a certain distance from the measured mean $\bar{x}$:

$$\text{Confidence interval} = \bar{x} \pm \frac{ts}{\sqrt{n}}$$

Where t is the student's t . taken from the table

Degree of freedom	Confidence Level (%)		
	90%	95%	99%
1	6.314	12.706	63.657
2	2.920	4.303	9.925
3	2.353	3.182	5.841
4	2.132	2.776	4.604
5	2.015	2.571	4.032
6	1.943	2.447	3.704
7	1.895	2.365	3.500
8	1.860	2.306	3.355
9	1.833	2.262	3.250
10	1.812	2.228	3.169
15	1.753	2.131	2.947
20	1.725	2.086	2.845

Validity of analytical results is determined by comparing mean of measured values with the true value, comparing means of measured results obtained by different methods, instruments, person etc. with each other and comparing individual differences between two methods, instruments, person etc. All these methods are based on comparison of means with student's t. A calculated t value is compared with the table t value with the same confidence level (95%) and same degree of freedom. Lower t calculated values than tabulated values means "no significant difference and absence of determinate errors".

Case 1. Comparing measured results with a "True" value:

$$t(\text{calculated}) = \frac{|m - \bar{x}|}{s} \sqrt{n}$$

Case 2. Comparing replicate measurements:

In this case we have two sets of replicates for the same analyte in the same method obtained by different methods or persons or instrument Etc. For the first set we calculated $\bar{x}_1$ and S_1 , for the second set we calculated

$\bar{x}_2$ and S_2 . Note that it's not necessary that $n_1=n_2$.

$$t \text{ (calculated)} = \frac{|\bar{x}_1 - \bar{x}_2|}{S_{pooled}} \sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

$$S_{pooled} = \sqrt{\frac{S_1^2(n_1 - 1) + S_2^2(n_2 - 1)}{n_1 + n_2 - 2}}$$

S_{pooled} is a pooled standard deviation making use of both sets of data, t (calculated) is compared with the tabulated one at $(n_1 + n_2 - 2)$ degree of freedom.

Case 3. Comparing individual differences:

In this case, we use two different methods to make single measurements on different samples.

$$t \text{ (calculated)} = \frac{|\bar{d}|}{S} \sqrt{n}$$

Where, $S_d = \sqrt{\frac{(d_i - \bar{d})^2}{n - 1}}$ and $\bar{d} = \sum_i \frac{d_i}{n}$

Q -Test for bad data:

Sometimes one datum is inconsistent with the remaining data. You can use the Q- test to help decide to retain or discard a suspect datum.

$$Q_{\text{calculated}} = \text{gap/range}$$

Where gap is the difference between suspect and next one and range is the difference between highest and lowest datum.

If $Q_{\text{calculated}} > Q_{\text{table}}$, the suspect point should be discarded, otherwise, it should be retained.

Number of observations	Q (90% confidence)
4	0.76
5	0.64
6	0.56
7	0.51
8	0.47
9	0.44
10	0.41

Procedure

Your instructor will supply you with two sets of objects. The first set is consisted from six 50-mL beakers and the second set is consisted from 10-mL, 25-mL, 50-mL, 100-mL and 250-mL beakers. Also two balances will be used in this experiment. Weigh each beaker from set one on balance one and balance two and record the weights on data sheet. Weigh each beaker from set two records the weights on data sheet.

Questions

1. Define the following terms: accuracy, precision, average, standard deviation, confidence interval, confidence level, true value, Q-test and students t.
2. The following result replicates for sodium in water 116.0, 97.9, 114.2, 106.8 and 108.3 ppm. Find the mean, standard deviation and 95% confidence interval of the mean, using the Q-test, decide whether the number 97.9 should be discarded.
3. A reliable assay shows that the ATP content is 111 $\mu\text{mol}/100\text{mL}$. You developed a new assay, which gave the following values for replicate analysis: 117, 119, 111, 115 and 120 $\mu\text{mol}/100\text{mL}$. Can you be 95% confident that your mean differs the reliable value?
4. Calcium in a mineral was analyzed five times by each of two methods. Are the mean values significantly different at the 95% confidence level?
Method 1. : 0.0271, 0.0282, 0.0279, 0.0271 and 0.0275.
Method 2. : 0.0271, 0.0268, 0.0263, 0.0274 and 0.0269.

Experiment (2) - Report sheet

Beaker	Wt. on balance 1	Wt. on balance 2
1		
2		
3		
4		
5		
6		
Average $\bar{x}$		
Standard deviation S		
Coefficient of variation CV		
Confidence interval		

t(tabulated) (95%) =

Case 1: Comparing measured results with a “True” value

True value (μ): $t_1(\text{calculated}) = \dots\dots$ $t_2(\text{calculated}) = \dots\dots$

Conclusion:

There is/ there is no significant difference between the true value (μ) and ($\bar{x}_1$)

There is/ there is no significant difference between the true value (μ) and ($\bar{x}_2$)

Case 2: Comparing replicate measurements

$S_{\text{pooled}} = \dots\dots\dots$ $t(\text{calculated}) = \dots\dots\dots$ $t(\text{tabulated}) (95\%) = \dots\dots\dots$

Conclusion:

There is/ there is no significant difference between $\bar{x}_1$ and $\bar{x}_2$

Rejection of bad data (Q -Test):

The suspect value: $Q_{\text{calculated}} = \dots\dots\dots$ $Q_{\text{table}} = \dots\dots\dots$

Conclusion:

The suspect value should be rejected/ retained.

Case 3: Comparing individual differences

Object	Balance 1	Balance 2	Difference
1			
2			
3			
4			
5			

Average deviation ($\bar{d} = \sum_i \frac{d_i}{n}$):

Standard deviation (S_d):

t (calculated) =

t (tabulated) (95%) =

Conclusion:

There is/ there is no significant difference between Balance₁ and Balance₂.

Experiment 3

Gravimetric Sulfate Determination

Purpose:

- To determine the percentage of SO_3 in an unknown sample.

Background

Gravimetric analysis is a quantitative chemical analysis used to determine the unknown concentration of one reactant (the analyte) by measuring the mass of precipitate formed when an excess of a second reactant (the precipitating reagent) is added to it. The solid precipitate is separated from the surrounding solution by filtering.

The steps required in gravimetric analysis:

1. Precipitation

Generally, precipitation reactions are slow, so that even when a precipitating reagent is added drop by drop to a solution of an analyte, some supersaturation is likely. The particle size of a precipitate varies inversely with the average relative supersaturation.

Relative supersaturation = $(Q - S)/S$

Q is the concentration of the solute at any instant and S is its equilibrium solubility.

When $(Q - S)/S$ is large: **nucleation** (few ions, atoms, or molecules come together to form a stable solid) may occur at a rate faster than crystal growth and the precipitate may behave as colloids (many small crystals) and pass through a filter.

When $(Q - S)/S$ is small: **crystal growth** (colloids come to gather to form a crystalline solid) may occur at a rate faster than nucleation producing fewer and larger crystals.

- If nucleation predominates, a precipitate containing a large number of small particles results
- If growth predominated, a smaller number of larger particles are produced.

To increase the rate of crystal growth, precipitating solutions may be heated. Because the solubility of most salts increases with increasing temperature, this will lower the relative degree of super saturation and slow the rate of nucleation. Also, the slow addition of precipitant with rapid mixing will avoid the occurrence of locally high concentrations.

Several steps are taken to control the particle size (keep Q low and S high during precipitation):

1. Precipitate from dilute solution. This keeps Q low.

2. Add dilute precipitating reagents slowly with effective stirring, this keeps Q low. Stirring prevents local excesses of the reagent.
3. Precipitate from hot solution. This increases S.
4. Precipitate at as low a pH as is possible because many precipitates are more soluble in acidic medium, and this slows the rate of precipitation

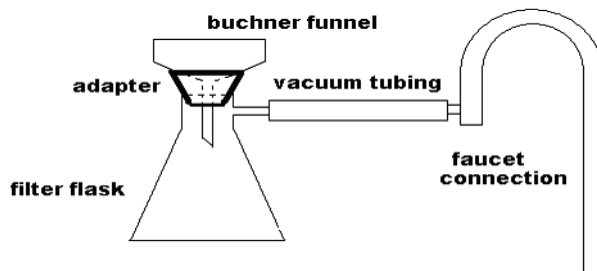
2. Digestion

The filterability of a coagulated colloid improves if it is allowed to stand for an hour or more in contact with the hot solution from which it was formed (the mother liquor). This helps in minimizing coprecipitation and obtaining larger crystals.

3. Filtration and Washing

The mixture is then filtered through a dry, weighed filter paper under vacuum

Vacuum filtration is carried out with a filter flask, a Buchner funnel, a rubber neoprene cone adapter, filter paper, a piece of vacuum tubing, and a source of vacuum (usually a connection on the side of an aspirator). For certain gravimetric analyses, substances besides the desired substances also get precipitated in small amounts (impurities), the phenomenon is called co-precipitation. It occurs when an unwanted ion or molecule becomes trapped in the precipitate. To avoid this, the glassware and precipitate are washed in small portions of distilled water



4. Drying or igniting

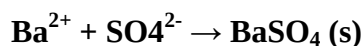
The filter paper and collected precipitate are dried in an oven at 105 °C for one hour.

5. Weighing

The dried filter paper and precipitate are cooled and weighed accurately. The mass of precipitate and stoichiometry are used to determine the concentration of the unknown analyte.

A sample containing sulfate is dried, weighed and dissolved in dilute HCl. Barium chloride solution is added in excess to precipitate barium sulfate, the acidic solution will not allow the barium salts of other anions such as phosphates and carbonates to coprecipitate.

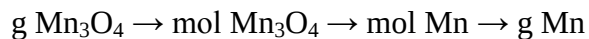
The precipitate is digested in the hot solution. The precipitate is then filtered through a filter paper. From the weight of the sample and weight of the precipitate, the percentage of sulfate in the sample is calculated.



Solved problems:

The manganese in 1.52 g of an ore is converted to Mn_3O_4 (228.8 g/mol), the weight of Mn_3O_4 is 0.126 g.

a) Calculate % Mn (54.94 g/mol).



$\text{weight Mn} = \text{weight Mn}_3\text{O}_4 * \text{Gravimetric factor (GF)}$

$\text{GF} = (\text{molecular weight Mn} / \text{molecular weight Mn}_3\text{O}_4) * (\text{mol Mn} / \text{mol Mn}_3\text{O}_4)$

$\text{GF} = (54.94 / 228.8) * (3/1)$

$\text{weight Mn} = 0.126 \text{ g} * \{(54.94 / 228.8) * (3/1)\} = 0.0908 \text{ g Mn}$

$\% \text{ Mn} = (0.0908 / 1.52) * 100\% = 5.97\%$

b) Calculate % Mn_2O_3 (157.9 g/mol) in the sample

$\% \text{ Mn}_2\text{O}_3 = 0.126 \text{ g} * \{(157.9 / 228.8) * (3/2)\} = 8.56\%$

Experiment (2) - Report sheet

Mass Unknown

Mass BaSO₄:

Gravimetric factor:

% SO₃ in the Unknown:

Calculation: